IRON HACK

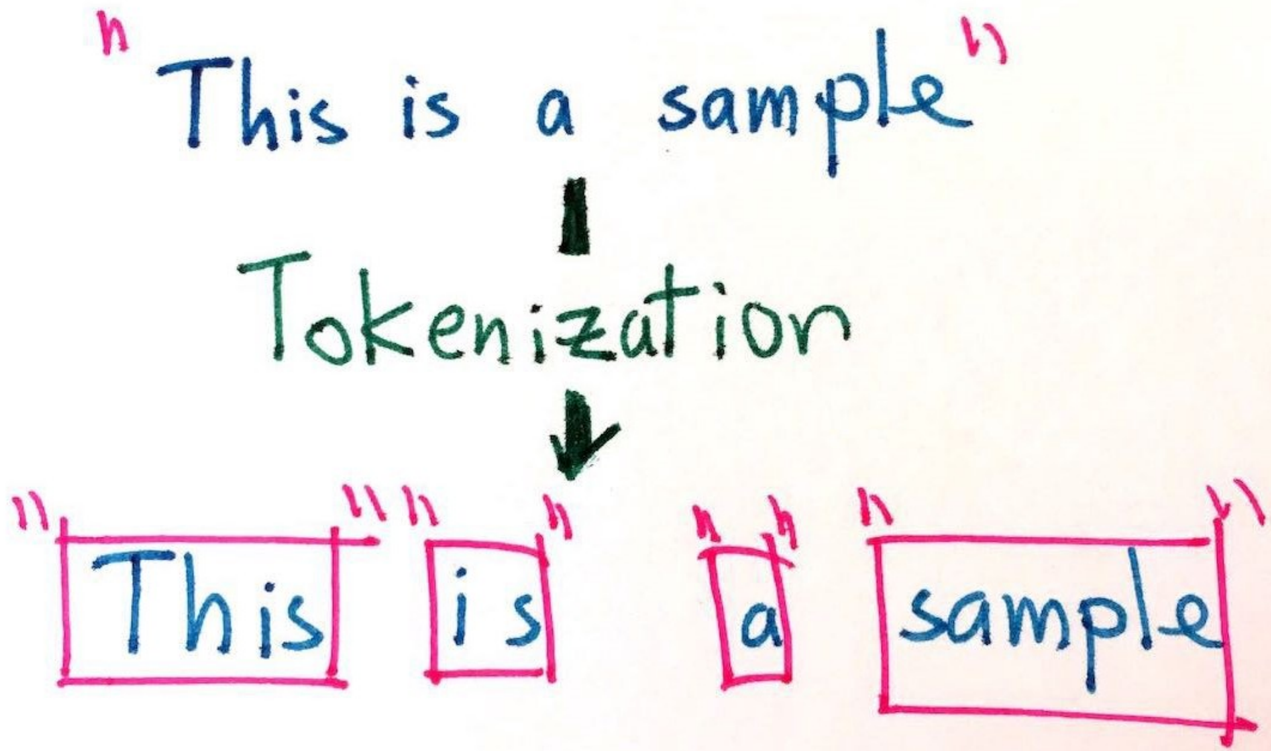
NATURAL LANGUAGE PROCESSING



Text Processing

TEXT DATA CLEANING TECHNIQUES

- Tokenization
- The process involves taking any text and splitting it into "tokens"
- Usually the tokens chosen are individual words but can be sentences as well.

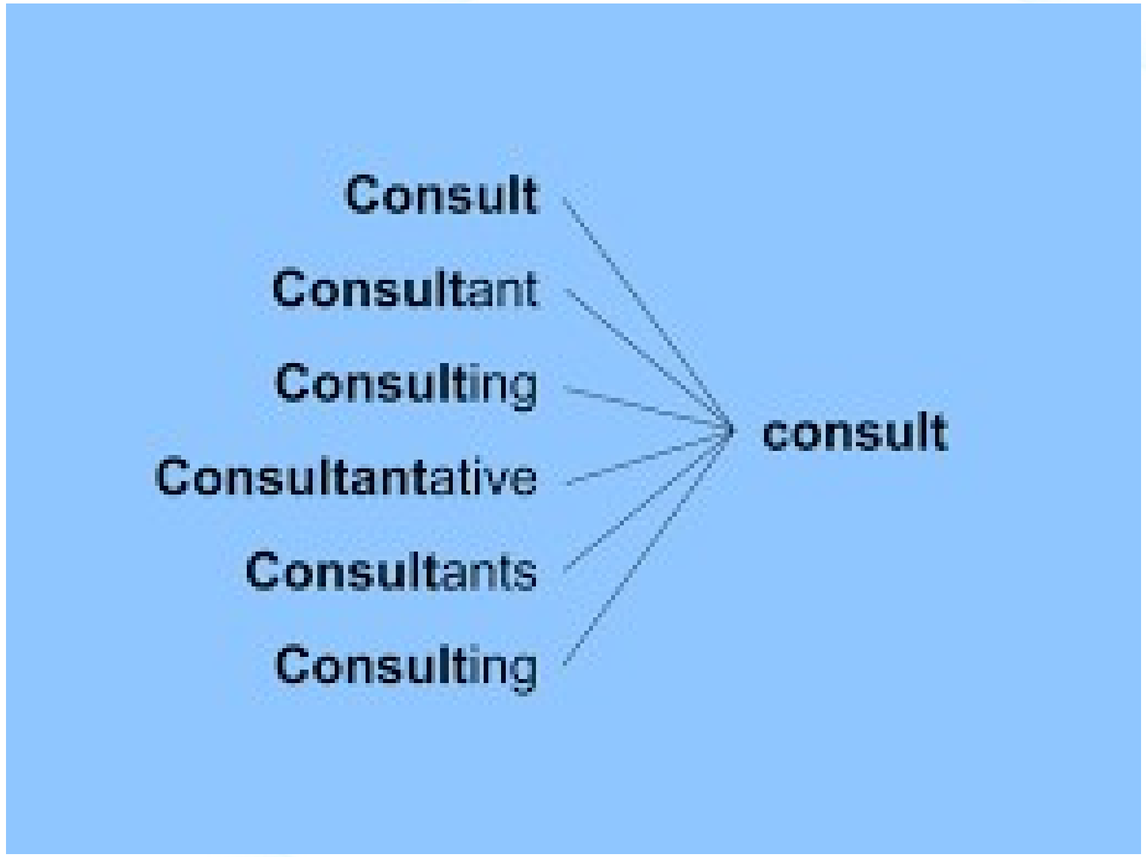


PROBLEMS: YOU CAN LOSE IMPORTANT MEANING LIKE "NEW YORK" GETS SPLIT INTO "NEW" AND "YORK" THUS LOSING THE GEOGRAPHICAL MEANING. A GOOD MACHINE LEARNING APPLICATION WOULD TAKE THIS INTO CONSIDERATION BY CONSIDERING THE RELATIVE FREQUENCYOF PAIRS OF WORDS AND PERHAPS KEEP THOSE PAIRS AS A SINGLE WORD

To collab...

TEXT DATA CLEANING TECHNIQUES

- Stemming
- Removing prefixes and suffixes in order to reduce synonym words into their common form
- This can lead to some problems of creating spurious relations
- Stripes -> Strip
- Strip -> Strip



To collab...

TEXT DATA CLEANING TECHNIQUES

lemmatization looks beyond word reduction and considers a language's full vocabulary to apply a morphological analysis to words. The lemma of 'was' is 'be' and the lemma of 'mice' is 'mouse'.



To collab...

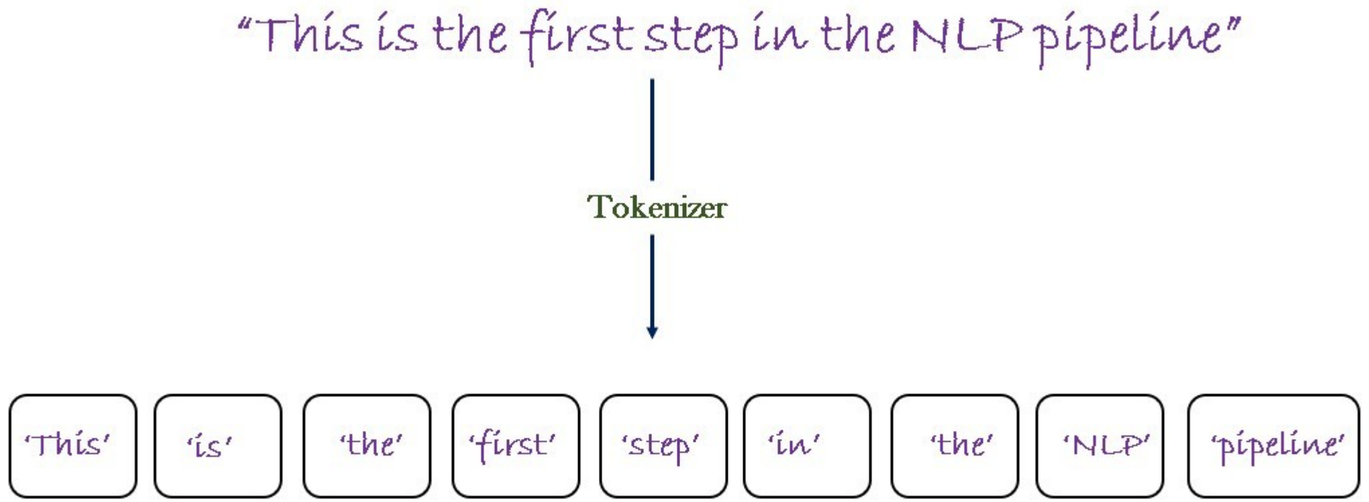
TEXT DATA CLEANING TECHNIQUES

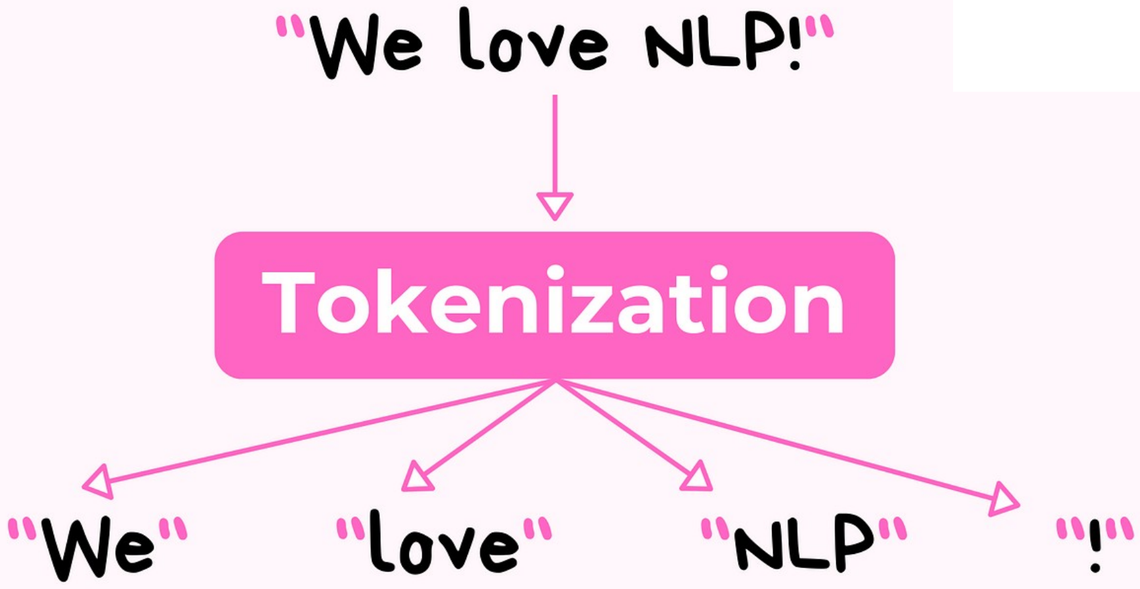
Removal of stop words

Stop words are common words which are usually removed to not be interpreted as noise

A	It	These
About	Its	They
Again	Itself	This
All	Just	Those
Almost	km	Thus
Also	Made	To
Although	Mainly	Upon
Always	Make	Use
An	May	Used
And	mg	Using
Another	Might	Various
Any	ml	Very
Are	mm	Was
As	Most	We
At	Mostly	Were

TOKENIZATION & STOPWORDS REMOVAL





Natural Language Processing with Python!

Stop Words

["This", "is", "a", "test"]

✔ X X ✔

Word-Embeddings

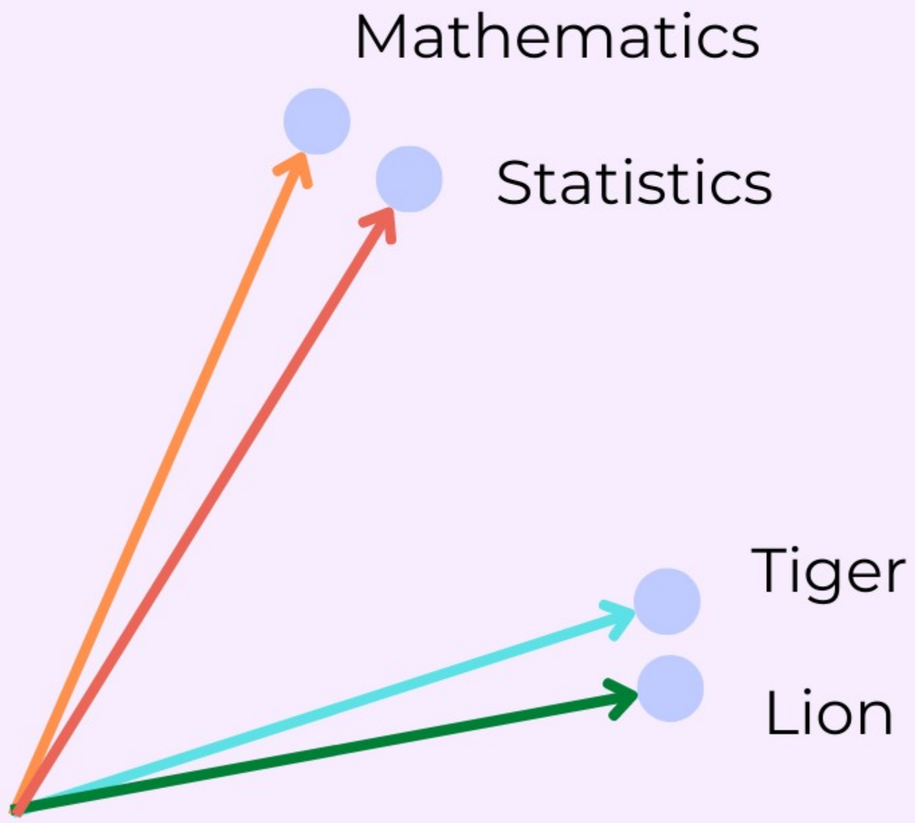
WHAT ARE WORD-EMBEDDINGS

Word embeddings are dense vector representations of words where similar words are located close to each other in the embedding space.

Each word is encoded as a point in a multi-dimensional space (e.g., 100, 200, or 300 dimensions).

- **Purpose**:
  - Embeddings translate the unstructured nature of language into numerical forms that machine learning models can process.
- **Dimensionality**:
  - Each dimension captures some aspect of a word’s meaning.
  - Higher-dimensional embeddings (e.g., 300) capture finer-grained relationships but require more computation.

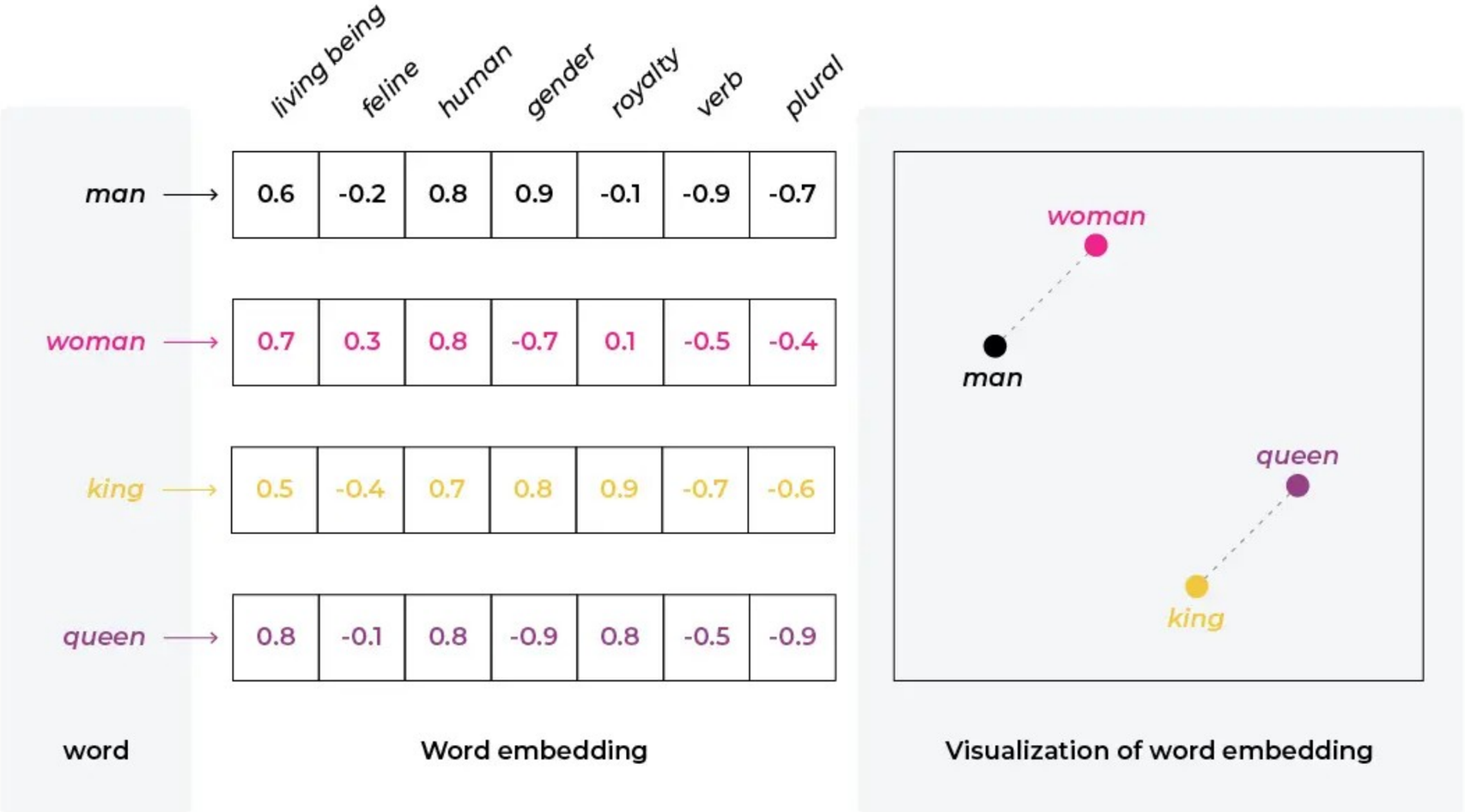
Word Embeddings



"Mathematics" is similar to "Statistics"

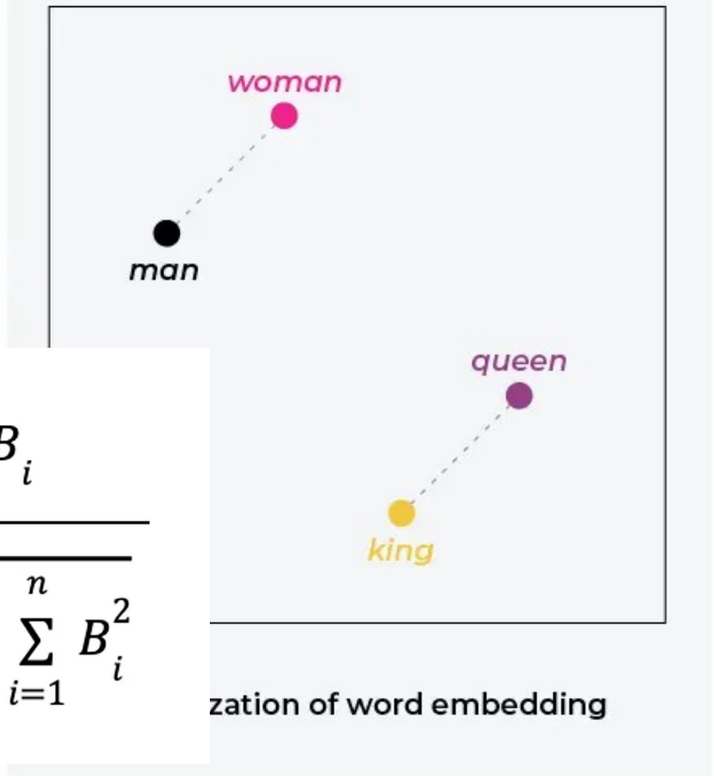
"Tiger" is similar to "Lion"

WHAT ARE WORD-EMBEDDINGS



WHAT ARE WORD-EMBEDDINGS

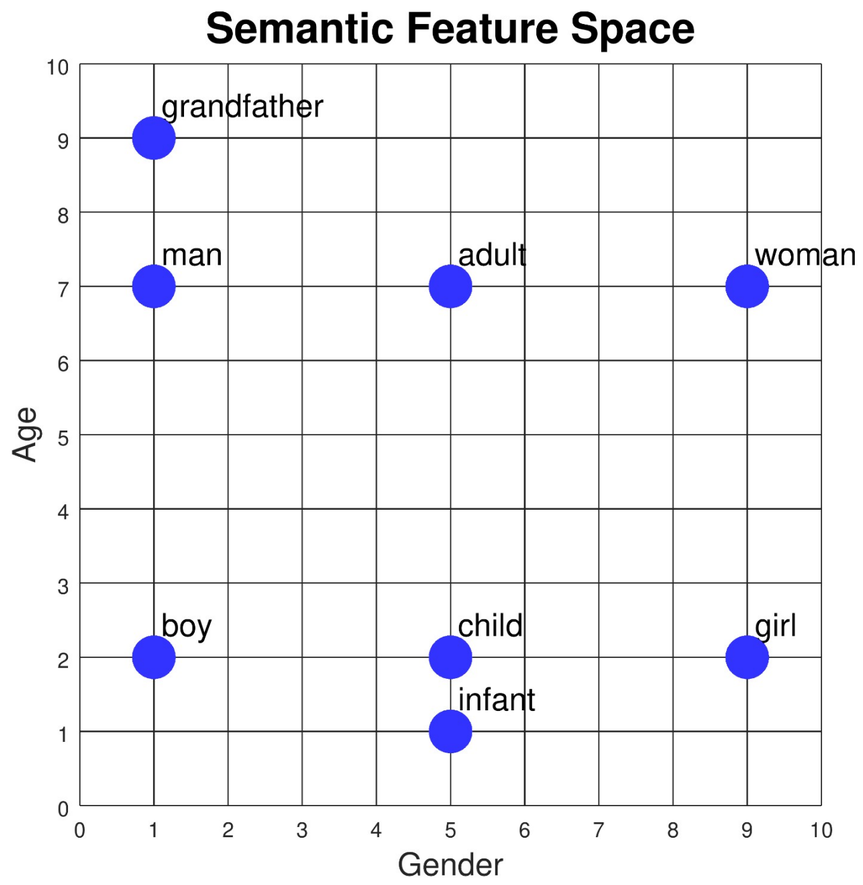
	living being	feline	human	gender	royalty	verb	plural
man →	0.6	-0.2	0.8	0.9	-0.1	-0.9	-0.7
woman →	0.7	0.3	0.8	-0.7	0.1	-0.5	-0.4

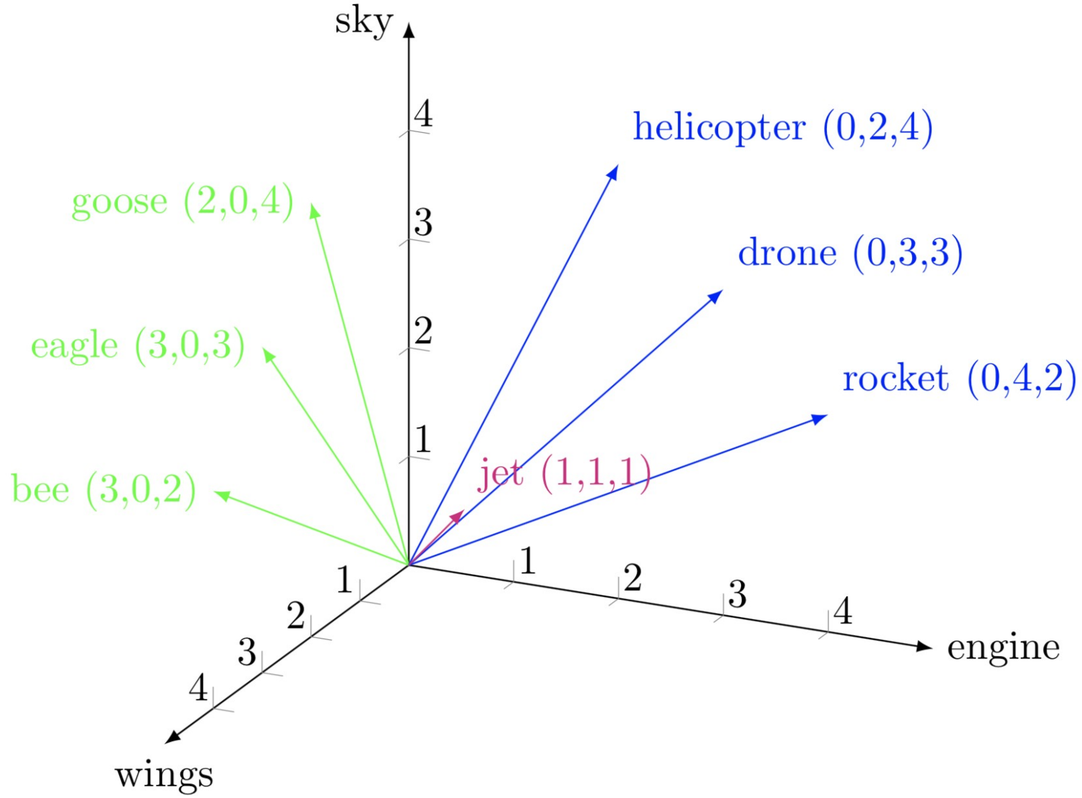


zation of word embedding

$$Cosine\ Similarity = \frac{\sum_{i=1}^{n} A_i \times B_i}{\sqrt{\sum_{i=1}^{n} A_i^2 \times \sum_{i=1}^{n} B_i^2}}$$

WHAT ARE WORD-EMBEDDINGS





Word-Embeddings

EMBEDDINGS

Pro : can be used pre-trained if your application is generic. For the vast majority of cases the word embeddings trained on Wikipedia are good enough

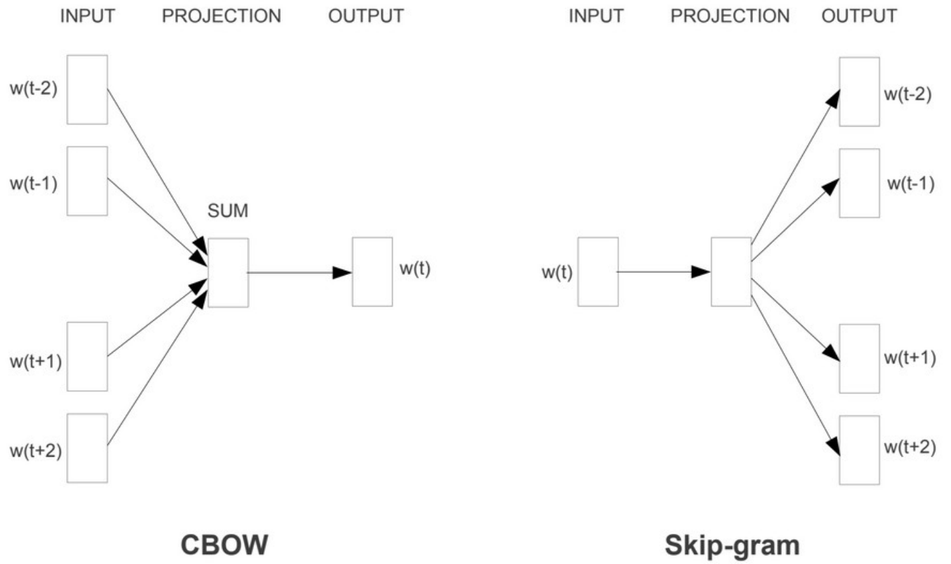
Pro: can be customized from a pre-learned model at a fraction of the cost. If you have a relatively modest corpus, you can piggyback on a pre trained model that already knows most of the smeantics of english

Con: can be quite difficult to train from scratch. Pretty much only google and a few other large data holders can create these models (or... OpenAI)

WORD EMBEDDINGS - WORD2VEC

Enriches the notion of autoencoder by considering not only the word but also the words that show up around it

Two versions, depending if it's more important to guess the word based on the context or predict the context based on the word



We have huge corpuses for training, essentially all of literature created so far

WORD EMBEDDINGS - BERT FAMILY

Word2Vec still associates the same vector to the same word

BERT - Bidirectional Encoder Representations from Transformers consider not only the word but also it's context *during encoding*, which means it can extract different meanings from the same word

"the man robbed a bank" vs "the bank of the river was flooded"

